Cog-Net: A Cognitive Network for Fine-Grained Ship Classification and Retrieval in Remote Sensing Images

Wei Xiong, Zhenyu Xiong, Libo Yao, and Yaqi Cui

***Abstract*— In light of the escalating volume of high-quality remote sensing ship images, the imperative task is to effectively classify and retrieve such images from extensive remote sensing archives. While prior research has yielded promising outcomes in ship classification, a comprehensive framework catering to both ship classification and retrieval remains absent. Additionally, prevailing studies neglect the critical aspect of model interpretability, merely furnishing predicted results devoid of a transparent reasoning process. This opacity, coupled with the high-stakes nature of outcomes, significantly impedes the safe utilization of these models. To address these problems, this article introduces the cognitive network (Cog-Net), an inherently interpretable model tailored for fine-grained ship classification and retrieval in remote sensing images. Cog-Net imitates the reasoning process employed by domain experts, navigating from perception to cognition during decision-making. The initial stage incorporates the causal multigrained feature learning (CMFL) module, mirroring the human perceptual process by identifying salient regions of a ship object within entire images as references for visual concept learning (VCL). Subsequently, the second stage introduces the VCL module and imitates the human cognitive process by learning basis visual concepts for each ship category and generating predictions through interpretable reasoning based on these basis visual concepts. Furthermore, to facilitate experimentation, a novel dataset, Fine-Grained Ship Remote Sensing Image Slices (FGSRSI-23), comprising 23 fine-grained ship subcategories, is constructed. Extensive experiments are conducted, encompassing our FGSRSI-23 dataset alongside two publicly available datasets, FGSC-23 and FGSCR-42. Results attest to the competitiveness of Cog-Net in both ship image classification and retrieval tasks, offering a transparent and interpretable reasoning process for predicted outcomes.**

***Index Terms*— Fine-grained ship classification, interpretable reasoning, remote sensing, ship retrieval.**

I. Introduction

IN RECENT years, with the vigorous development of space-based ocean monitoring systems, satellite imaging technology has garnered widespread attention as a crucial means of early warning and detection. The tasks of identifying and retrieving ship targets in remote sensing imagery are of paramount importance in both the military and civilian domains. However, given the highly dynamic nature of maritime environments, the extensive variety of ship types, and the complex background information present in remote sensing images, identifying ship types and simultaneously finding ship target images of interest to users is exceptionally challenging. Thanks to the powerful feature representation capabilities of deep learning, some studies successfully distinguish the remote sensing ship images in a coarse-grained manner [1], [2], [3].

With the rapid advancement of satellite imaging technology, there is a growing proliferation of remote sensing imagery available to individuals, coupled with an escalation in spatial resolution. Zhang et al. [4] and Di et al. [5] have leveraged this development to establish fine-grained remote sensing ship image datasets, FGSC-23 and FGSCR-42, respectively. In contrast to general coarse-grained classification tasks, fine-grained remote sensing ship image classification necessitates the differentiation of subordinate ship categories. The challenge lies in discerning subtle visual distinctions among ship images that, despite the overall similarity, exhibit nuanced variations. As depicted in Fig. 1, the various subcategories within the destroyer category from our newly constructed dataset, FGSRSI-23, demonstrate high interclass similarity, posing a classification difficulty. To address these problems, some studies attempt to design deep networks to classify ship images in a fine-gained manner based on the above-mentioned two datasets [6], [7], [8], [9]. For instance, Chen et al. [6] incorporate a dual-branch network and an integration module to decorrelate images and group them into subclasses, addressing the challenges of intraclass dissimilarity and interclass similarity among ship subclasses and improving classification performance. Yi et al. [7] obtain highly discriminative features by incorporating a miner for feature purification and a refiner for data augmentation, enhancing the accuracy of target classification. Zhu et al. [8] address the challenges of high interclass similarity and large intraclass differences by incorporating a prototype separability module for interclass prototype distribution update and a meta-network based on a foreground enhancement mechanism for more representative intraclass prototypes. However, despite the fact that the above-mentioned studies already achieve satisfactorily in fine-grained ship classification tasks, they also face limitations in effectively extending to image retrieval tasks. The core objective of fine-grained remote sensing ship image retrieval is, for a given query ship image, to search for the most

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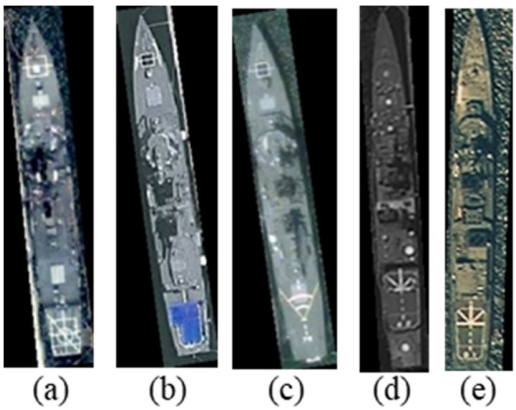

Fig. 1. Some ship images belonging to the destroy category from our proposed dataset FGSRSI-23. (a) ABD. (b) ATD. (c) KOD. (d) ASD. (e) AKD.

semantically similar image from a large-scale ship image database by measuring the distance in the feature space. Notably, the learned features tend to be separable, rather than purely discriminative, which is inconsistent with the training purpose of the classification task. One of the motivations of the proposed study is to construct a network that can simultaneously process fine-grained ship classification and retrieval tasks.

Beyond the model's discriminative capacity, interpretability assumes significant importance. The prevalent deep learning techniques primarily aim at training networks to adapt and align with training datasets, neglecting to provide a coherent explanation for the ultimate decision outcomes. Consequently, placing our trust entirety of the final outcomes generated by black-box models becomes a highly risky task. To clarify where the remote sensing image regions that the deep model focuses on, Huang et al. [10] provide visual interpretations by generating heatmaps of remote sensing scene images. Guo et al. [11] enhance low-level information through channel saliency and gradient strategy, yielding stable saliency maps for remote sensing scene images. However, these class activation map-based methods fail to clarify the decision logic behind each filter of the deep network simply by visualizing the highest activation of the whole remote sensing images, which will result in interpretative chaos. For the image classification, these models fail to explain why Fig. 2(a) is categorized as an amphibious assault ship through a generated heatmap. In Fig. 2(b), for image retrieval, the heatmaps alone cannot elucidate why A is similar to B but dissimilar to C. Seeking to clarify the network's reasoning process, Xiong et al. [12], [13] adopt the filter aggregation mechanism where each filter group represents a specific part of ship images, quantifying the contribution of each part to the final prediction. However, it is crucial to underscore that these approaches furnish posthoc explanations, offering only an approximate understanding and indirect interpretation. Such methods may not align with human cognitive decision-making, potentially resulting in discrepancies and misleading interpretations [14]. The major motivation of this study is to provide interpretable results that align with the human reasoning process for the fine-grained ship classification and retrieval tasks.

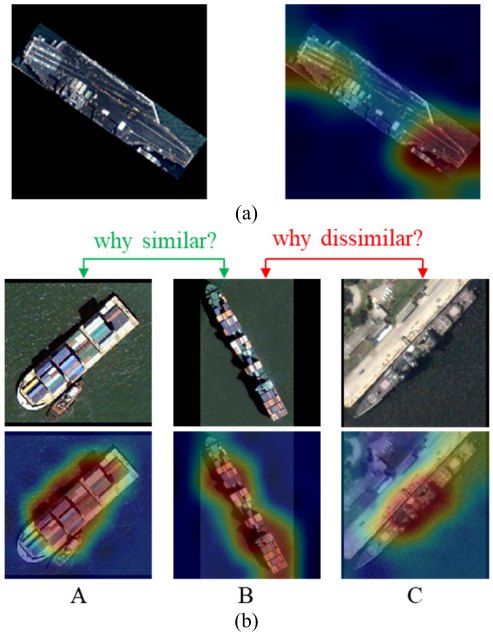

Fig. 2. (a) Image of the NAC category from our FGSRSIS-23 dataset, along with its corresponding heatmap. (b) Two images of the Container Ship category (A, B) and an image of the Cruise category (C) from FGSC-23 dataset, accompanied by their respective heatmaps.

When performing ship image processing tasks, domain experts tend to focus on distinct regions of a given ship image and use them for reasoning to obtain prediction results, which involves a process from a person's ***primary perception*** to ***advanced cognition***. We hope the proposed deep model can imitate this decision-making process by reasoning several key parts of ship images. To formulate an inherently interpretable network aligned with human cognition, we propose the cognitive network (Cog-Net) for fine-grained classification and retrieval of remote sensing ship images, elucidating the reasoning process. In the first stage, we imitate the human ***perceptual*** process, searching for salient regions of a ship object within entire images for reference in visual concept learning (VCL). The **causal multigrained feature learning** (CMFL) module is proposed, extracting global features from the last convolutional layer of the backbone through global average pooling (GAP) for each channel's average value. In the causal local branch of CMFL, we generate attention maps to represent distinct parts of ship images. Acknowledging potential spurious relationships between salient regions and predicted results, we explore intrinsic relationships to ensure confounder-free local features, constructing a causal graph to instill counterfactual thinking akin to human cognition. Subsequently, the causal multigrained feature is derived by merging global and causal local features. In the second stage, mirroring the human ***cognitive*** process, the network learns basis visual concepts for each ship category and produces predictions through visual reasoning based on these acquired concepts. In our proposed VCL module, we measure the similarity between causal multigrained features and several basis visual concepts by calculating the correlation score. In the training process, well-designed constraints are applied to guarantee diversity and discriminability of learned basis

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visual concepts. After that, we concatenate the correlation scores to generate a feature vector for prediction. To facilitate both image classification and retrieval tasks, we employ sphereface [15] and focal loss [16] as the task loss functions to optimize the proposed network.

Furthermore, given the inadequacy of the currently available public datasets for remote sensing ship images, namely, FGSC-23 and FGSCR-42, in verifying the discriminability and interpretability of our model, we construct a new dataset, Fine-Grained Ship Remote Sensing Image Slices (FGSRSI-23). Extensive experiments are conducted based on the aforementioned two public datasets and our newly introduced FGSRSI-23 dataset. The qualitative and quantitative results demonstrate the proposed network outperforms the majority of existing state-of-the-art methods in both ship image classification and retrieval tasks. Moreover, it offers an interpretable reasoning process for the predicted results.

The main contributions of this article are highlighted as follows.

1) We propose Cog-Net, an inherently interpretable network for fine-grained ship classification and retrieval in remote sensing images, which is the first work that imitates domain expert reasoning.
2) The proposed CMFL module imitates the human perceptual process to find salient features in ship images. The proposed VCL module imitates the human cognitive process, learning basis visual concepts for decision-making.
3) We release a new public dataset, FGSRSI-23, comprising nine primary and 23 subcategories for comprehensive evaluation of Cog-Net's effectiveness.
4) The Cog-Net is an interpretable network with robust discriminability for ship image classification and retrieval. Its versatility makes it applicable in military and civilian domains, offering promising prospects.

The subsequent sections of this article are structured as follows. Section II encompasses a comprehensive review of the existing literature pertaining to fine-grained remote sensing image classification, retrieval, and interpretable feature representation. Section III presents the detailed architecture of the proposed network. Section IV enumerates the experimental outcomes and associated analyses. Finally, Section V presents the conclusions drawn from this article.

II. Related Work

A. Fine-Grained Remote Sensing Images Classification

Fine-grained image classification has garnered considerable attention within the field of computer vision. In the realm of remote sensing, early studies have predominantly concentrated on finely distinguishing remote sensing scenes [17], [18], [19], [20], [21]. Lv et al. [17] extract both local and global features, utilizing a novel recurrent neural network-based attention module for spatial context information to address the challenge of nondiscriminative feature representations. Wang et al. [18] employ a global-local two-stream architecture to locate important regions within the images using only image-level labels, improving classification performance in the presence of large-scale variations in remote sensing images. Li and Wang [21] address intraclass differences and interclass similarity in complex background and variable scale scenarios by obtaining precise local key areas and fusing multilayer semantic information. These methods are proposed to extract features at various levels from the entire images, neglecting to emphasize the discriminative features of objects within them. The inclusion of complex background information and subtle differences among different subcategories hampers the ability of these methods to effectively classify objects in the images.

As the spatial resolution of remote sensing images captured by optical satellites has improved, recent studies have focused on fine-grained ship image classification [6], [7], [8]. Zhao et al. [22] introduce a novel pipeline to address the challenge of blurred or missing object details, with a focus on integrating super-resolution strategies, feature balance, and iterative interaction mechanisms to improve performance. Li and Bian [23] effectively address the challenge of limited ship images by reconstructing query features from support features and using foreground weights to improve model generalization. Li et al. [24] address the challenges of few-shot ship classification and the significance of feature map channels by assigning different weights to channels based on their discriminative details, leading to improved classification accuracy. However, the aforementioned methods treat deep networks as a "black box" focused solely on enhancing performance through constraints and data augmentation strategies, overlooking the interpretability of the deep model. In this study, we introduce an inherently interpretable network that imitates domain expert reasoning. The proposed method not only classifies ship images in a fine-grained manner but also elucidates the network's reasoning process by outputting key elements of ship objects.

B. Fine-Grained Remote Sensing Images Retrieval

The objective of fine-grained remote sensing image retrieval is to search for images containing relevant information content from a large amount of gallery remote sensing images, given a specific query image. Existing studies have predominantly concentrated on retrieving remote sensing scene images. For example, Xiong et al. [25] focus on salient features while suppressing others and use a multitask learning network structure to make learning-based features more discriminative. By designing aggregating local descriptors, Imbriaco et al. [26] extract discriminative and attentive features and accurately retrieve the remote sensing scene images. Liu et al. [27] combine a deep feature learning model and an adversarial hash learning model to learn dense features for retrieval precision. Zhao et al. [28] incorporate multiscale residual blocks to capture fine-grained multiscale features and employ multicontext attention hash codes to enhance the perceptual field and reduce interference from irrelevant information. While hash-based methods can enhance the efficiency of scene image retrieval, the generation of binary codes leads to a loss of feature information, consequently diminishing retrieval accuracy in fine-grained ship retrieval tasks.

For fine-grained image retrieval of remote sensing ship images, two learning-based networks are proposed to retrieve

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ship images from multispectral and panchromatic remote sensing image modalities [13], [29]. Guo et al. [29] utilize a vision transformer as the feature extractor and employ a two-step feature learning strategy to transfer knowledge between modalities while addressing the challenges of complex geometric features and modality gaps, ultimately improving retrieval precision. Xiong et al. [13] propose a siamese pipeline to extract discriminative features of ship images and provide reasonable interpretations for the retrieval result. Nevertheless, both of these two approaches are tailored for retrieving cross-modality ship images within only six ship categories, including destroyers, combat boats, container ships, and so on. This design imposes limitations on their effectiveness in distinguishing subcategory ship images. In this study, we address this limitation by capturing key regions of ship objects through the extraction of causal multigrained features. Additionally, we introduce a task loss function to ensure the discriminability of the learned ship image features.

C. Interpretable Feature Representation

Generating interpretable feature representations typically involves visualizing the network's focal points. This can be achieved by computing the contribution of each image pixel to the ultimate outcome and generating corresponding heatmaps. Zhou et al. [30] introduce a class activation mapping (CAM) technique that adds a fully connected (FC) unit to the last layer of deep networks to identify the maximum activation fields of latent layers. Building on CAM, gradient-weighted CAM (GradCAM) [31] is proposed, utilizing class-specific gradient information to generate attention maps for visualization. Chattopadhay et al. [32] further develop GradCAM++, which employs the weights of the last convolutional layer to calculate scores for each class, accommodating explanations for multiple objects within a single image. Dosovitskiy and Brox [33] apply an upconvolutional network to provide an interpretable visual representation for both manually designed and learned features. Zhou et al. [34] demonstrate that CNNs possess the capability to autonomously detect objects in images during the training process.

Nonetheless, most interpretable approaches primarily focus on natural image processing, leading to less than satisfactory visualization outcomes for remote sensing images. This is attributed to the smaller object size and more intricate background information in remote sensing images compared with natural images. To accurately visualize critical regions in remote sensing images, a bidirectional gradient verification technique [35] is introduced to generate internal feature maps for correcting false visualizations. Huang et al. [10] utilize an image reconstruction technique to aid the interpretable model in precisely identifying significant objects in remote sensing images. Guo et al. [11] improve low-level information in the feature map by implementing channel saliency and gradient strategies, yielding stable saliency maps for remote sensing scene images. Xiong et al. [12], [13] incorporate a filter aggregation mechanism, ensuring that each filter group represents a specific portion of ship images and quantifies the contribution of each part to the final prediction.

It is worth mentioning that the aforementioned approaches offer posthoc interpretations for the trained networks, which are merely approximate and indirect interpretations. These interpretations might not align with human cognitive decision-making, potentially leading to erroneous interpretations. To develop a model that aligns with the human reasoning process, Chen et al. [36] propose a novel case-based reasoning strategy to learn some prototypes by focusing on different parts of objects and using them to clarify the reasoning process of the network's prediction. Followed by this framework, Wang et al. [37] bridge high-level input patches and output categories by introducing a plug-in transparent embedding space. Rymarczyk et al. [38] introduce the prototypical part-shared network that uniquely fosters prototype sharing across categories. Nauta et al. [39] merge prototype learning and decision trees to establish a general interpretable deep learning approach for fine-grained image identification. Rymarczyk et al. [40] propose a prototype-based model that tackles the drawbacks of existing approaches by reusing prototypes across classes. Donnelly et al. [41] combine deep learning capabilities with case-based reasoning, improving the accuracy and richness of explanations. Based on the vision transformer backbones, Xue et al. [42] and Lin and Wang [43] learn descriptive and discriminative prototype representations based on the transformer architecture for deep learning models, providing interpretability without requiring ground truth attribute supervision. Moreover, recent studies have successfully applied case-based reasoning to the various fields, including medical science [44], [45], [46], incremental continual learning [47], semantic segmentation [48], sequence learning [49], [50], [51], video analysis [52], [53], and zero-shot learning [54]. Inspired by these studies, we imitate the process from human primary perception to advanced cognition for fine-grained ship classification and retrieval in remote sensing images.

III. Proposed Method

A. Overview of the Method

Assuming access to an optical remote sensing ship image dataset $\{\mathbf{X}, Y\}$, where $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{N} \in \mathbb{R}^{H\times W\times 3}$ and $Y = \{y_i\}_{i=1}^{N} \in \mathbb{R}$ denote the N input images and their associated labels, respectively. As depicted in Fig. 3, for a given input image $\mathbf{x}_i$, we employ a convolutional backbone to extract high-level features $\mathbf{F}_h \in \mathbb{R}^{H\times W\times D}$ following a series of convolution and pooling operations, where W, H, and C represent the weight, height, and number of channels of feature maps, respectively. Subsequently, $\mathbf{F}_h$ is fed into the CMFL module to obtain the global features $\mathbf{F}_g \in \mathbb{R}^{1\times 1\times D}$ and causal local features $\mathbf{F}_c \in \mathbb{R}^{M\times 1\times 1\times D}$. The details of CMFL are described in Section III-B. Afterward, the fused causal multigrained features $\mathbf{F}_f \in \mathbb{R}^{M\times 1\times 1\times 2D}$ are generated by concatenating $\mathbf{F}_g$ and $\mathbf{F}_c$. Utilizing the FC layer, we reduce the dimensionality of $\mathbf{F}_f$ and employ it as input for the VCL module. During the training stage, VCL aims to generate basis visual concepts $\mathbf{v} \in \mathbb{R}^{M\times 1\times 1\times D}$ for each ship category, providing explainable evidence for downstream tasks. This is achieved by computing the correlation score between $\mathbf{F}_f$

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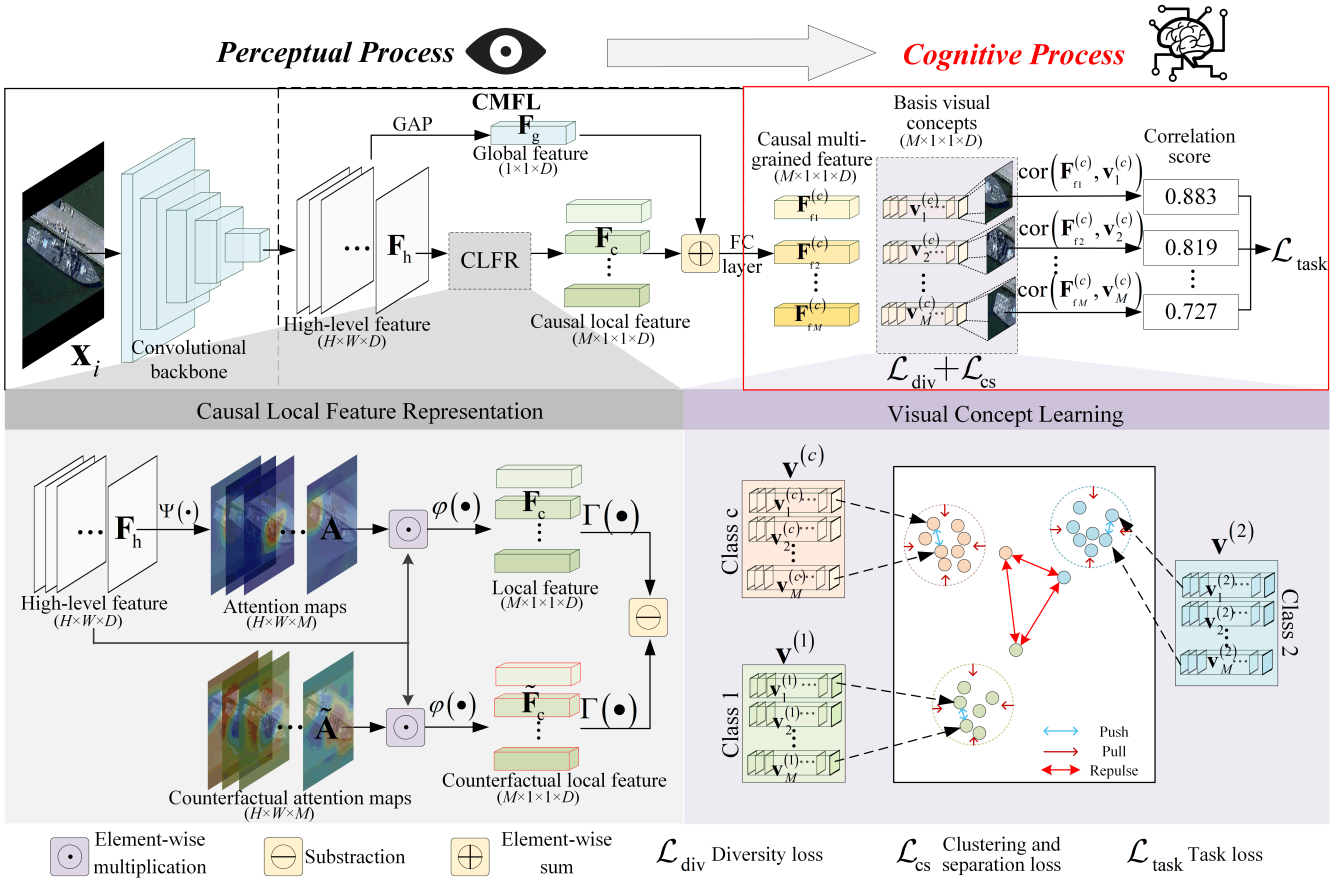

Fig. 3. Overall framework of the proposed Cog-Net. The Cog-Net imitates the reasoning process of domain experts, making decisions through perception to cognition. In the perceptual process, based on the high-level feature, Cog-Net fuses the causal local feature obtained by the CLFR branch and the global feature to generate the causal multigrained feature. In the cognitive process, we measure the similarity between causal multigrained features and several basis visual concepts by calculating the correlation score. We use $\mathcal{L}_{\text{div}}$ and $\mathcal{L}_{\text{cs}}$ to guarantee the diversity and discriminability of the learned basis visual concept. After that, we concatenate the correlation scores to generate feature vector for prediction.

and $\mathbf{v}$, which is then concatenated to form a feature vector for prediction. In the testing stage, the correlation score between the causal multigrained feature of test images and trained basis visual concepts is calculated, and these are employed in the classification or retrieval predictions. The details of VCL are presented in Section III-C.

B. Causal Multigrained Feature Learning

In the initial stage, leveraging the high-level feature $\mathbf{F}_h$, we extract the global feature from the entire image to capture comprehensive and coarse-grained information. Specifically, GAP is employed to compute the average value of each channel, with the output of the pooling layer being regarded as a global feature $\mathbf{F}_g \in \mathbb{R}^{1\times1\times D}$ for the images.

Building upon the success of our prior investigation [12] into fine-grained remote sensing ship image classification, the second stage employs counterfactual leaning to obtain causal local feature representation (CLFR). Traditional deep networks train models to match and fit observed data, focusing solely on exploring associations and disregarding causality between attention regions and predicted results. The spurious associations arising from attention biases can restrict the network's performance. Our module imitates the human perceptual process, incorporating counterfactual thinking during decision-making. As shown in Fig. 4, we construct a causal graph to demonstrate the counterfactual intervention technique. For the input images $\mathbf{X}$ and the attention maps $\mathbf{A}$ generated by attention models, predicted labels $Y(\mathbf{A})$ are affected by both $\mathbf{X}$ and $\mathbf{A}$. To implement the counterfactual intervention technique, several counterfactual attention maps $\tilde{\mathbf{A}}$ are generated, and their corresponding predicted labels $Y(\tilde{\mathbf{A}})$ are obtained. The true effect of attention regions on predicted labels Y_{true} is determined by calculating the difference between $Y(\mathbf{A})$ and $Y(\tilde{\mathbf{A}})$.

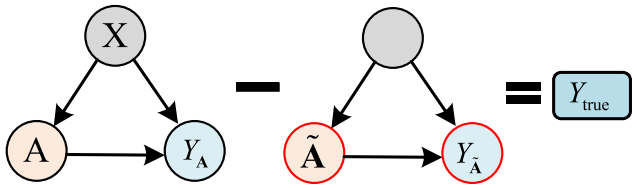

Fig. 4. Causal graph for the counterfactual intervention.

Based on the counterfactual intervention illustrated in Fig. 4, we elaborate on the CLFR module presented in Fig. 3. Given the high-level feature $\mathbf{F}_h$, we perform a 2-D convolutional

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function to generate several attention maps $\mathbf{A} = \Psi(F^{l-1})$, $\mathbf{A} = \{\mathbf{A_1}, \mathbf{A_2}, \ldots, \mathbf{A}_M\} \in \mathbb{R}^{W\times H\times M}$. Here, Ψ represents the 1×1 kernel size 2-D convolutional function. M is the number of attention maps. Notably, $\mathbf{A}_m \in \mathbf{A}$ signifies a specific region or visual pattern of the ship object the network concentrates on. In the subsequent step, we elementwise multiply high-level feature $\mathbf{F}_h$ by attention maps $\mathbf{A}$ and employ GAP to generate the local feature representation $\mathbf{F}_c = \{\mathbf{F}_{c1}, \mathbf{F}_{c2}, \ldots, \mathbf{F}_{cM}\} \in \mathbb{R}^{1\times1\times D\times M}$. This can be formulated as follows:

$$\mathbf{F}_c = \begin{pmatrix} \varphi(\mathbf{F}_h \odot \mathbf{A}_1) \\ \varphi(\mathbf{F}_h \odot \mathbf{A}_2) \\ \cdots \\ \varphi(\mathbf{F}_h \odot \mathbf{A}_M) \end{pmatrix} = \begin{pmatrix} \mathbf{F}_{c1} \\ \mathbf{F}_{c2} \\ \cdots \\ \mathbf{F}_{cM} \end{pmatrix} \tag{1}$$

where $\odot$ refers to the elementwise multiplication between two tensors. φ stands for the GAP operation. We aggregate the local feature by summing operation and then feed it into a classifier that consists of an FC layer and a batch normalization layer. We utilize cross-entropy as our loss function, represented by $\mathcal{L}_{\text{self}}$. The output of the classifier $\Gamma(\cdot)$ is expressed as follows:

$$Y_{\mathbf{A}}(\mathbf{F}_h) = \Gamma\left(\sum_{m=1}^{M} \mathbf{F}_{\text{cm}}\right) = \Gamma\left(\sum_{m=1}^{M} \varphi(\mathbf{F}_h \odot \mathbf{A}_m)\right). \tag{2}$$

To implement the counterfactual intervention technique, following the practice in [12], we generate some counterfactual attention maps $\tilde{\mathbf{A}}$, where the attention map vectors are assigned a uniform distribution with a zero mean. The classifier's output after performing the counterfactual operation is expressed in the following:

$$Y_{\tilde{\mathbf{A}}}(\mathbf{F}_h) = \Gamma\left(\sum_{m=1}^{M} \tilde{\mathbf{F}}_{\text{cm}}\right) = \Gamma\left(\sum_{m=1}^{M} \varphi\left(\mathbf{F}_h \odot \tilde{\mathbf{A}}_m\right)\right). \tag{3}$$

The direct effects can be expressed by computing the difference between the observed prediction $Y_{\mathbf{A}}(\mathbf{F}_h)$ and its counterfactual results $Y_{\tilde{\mathbf{A}}}(\mathbf{F}_h)$

$$Y_{\text{true}} = Y_{\mathbf{A}}(\mathbf{F}_h) - Y_{\tilde{\mathbf{A}}}(\mathbf{F}_h) \tag{4}$$

where Y_{true} denotes the direct effects of learned attention on the predicted results. In this module, the causal local loss function $\mathcal{L}_{\text{cau}}$ is employed to guide the causal local feature learning procedure, as shown in the following expression:

$$\mathcal{L}_{\text{cau}} = \mathcal{L}_{\text{ce}}(Y_{\text{true}}, y) + \mathcal{L}_{\text{self}} \tag{5}$$

where y indicates the ground truth label, and $\mathcal{L}_{\text{ce}}$ indicates the cross-entropy loss. The causal multigrained feature $\mathbf{F}_f = \{\mathbf{F}_{f1}, \mathbf{F}_{f2}, \ldots, \mathbf{F}_{fM}\} \in \mathbb{R}^{1\times1\times2D\times M}$ is obtained by concatenating the global feature $\mathbf{F}_g \in \mathbb{R}^{1\times1\times D}$ to each causal local feature $\mathbf{F}_c = \{\mathbf{F}_{c1}, \mathbf{F}_{c2}, \ldots, \mathbf{F}_{cM}\} \in \mathbb{R}^{1\times1\times D\times M}$. This can be formulated as follows:

$$\mathbf{F}_f = \begin{pmatrix} \mathbf{F}_{c1} \oplus \mathbf{F}_g \\ \mathbf{F}_{c2} \oplus \mathbf{F}_g \\ \cdots \\ \mathbf{F}_{cM} \oplus \mathbf{F}_g \end{pmatrix} = \begin{pmatrix} \mathbf{F}_{f1} \\ \mathbf{F}_{f2} \\ \cdots \\ \mathbf{F}_{fM} \end{pmatrix}. \tag{6}$$

C. Visual Concept Learning

In the VCL model, we aim to generate M basis visual concepts $\mathbf{v}^{(c)} = \{\mathbf{v}_1^{(c)}, \mathbf{v}_2^{(c)}, \ldots, \mathbf{v}_M^{(c)}\} \in \mathbb{R}^{1\times1\times D\times M}$ for each ship image category c, where $\mathbf{v}_m^{(c)} \in \mathbb{R}^{1\times1\times D}$ denotes one of the basis visual concept of ship category c. For all ship categories, there exists an $M \times C$ basis visual concept $\mathbf{V} = \{\mathbf{v}^{(1)}, \mathbf{v}^{(2)}, \ldots, \mathbf{v}^{(C)}\}$.

To accomplish this, given the generated causal multigrained feature $\mathbf{F}_f \in \mathbb{R}^{1\times1\times D\times M}$ after dimensionality reduction, we perform a correlation function to measure the similarity between $\mathbf{F}_f$ and $\mathbf{V}$, which is given by

$$\begin{aligned} \text{cor}\left(\mathbf{F}_{fm}^{(c)}, \mathbf{v}_m^{(c)}\right) &= \frac{1}{2}\left(\rho_{\mathbf{F}_{fm}^{(c)}\mathbf{v}_m^{(c)}} + 1\right) \\ &= \frac{1}{2}\left(\frac{\text{cov}\left(\mathbf{F}_{fm}^{(c)}, \mathbf{v}_m^{(c)}\right)}{\sqrt{\text{var}\left(\mathbf{F}_{fm}^{(c)}\right)}\sqrt{\text{var}\left(\mathbf{v}_m^{(c)}\right)}} + 1\right) \end{aligned} \tag{7}$$

where $\mathbf{F}_{fm}^{(c)}$ and $\mathbf{v}_m^{(c)}$ denote the mth causal multigrained feature and basis visual concept of category c. $\text{cor}(\cdot)$ indicates the correlation function. $\rho_{\mathbf{F}_{fm}\mathbf{v}_m^{(c)}}$ is the Pearson correlation coefficient (PCC).

To align with human cognitive processes, different basis visual concepts should uniquely represent different parts of ship images even though they belong to the same category. For instance, a ship image classified as an "Atago-class Destroyer (ATD)" should exhibit several distinct basis visual concepts, such as "foredeck region" and "superstructure." To ensure diverse and rich basis visual concepts for predicting results, it is essential to prevent overlap between different visual concepts. Inspired by BarlowTwins [55], we employ a diversity loss function, aiming to bring the correlation matrix between basis visual concepts of the same category close to the identity matrix. This ensures that each basis visual concept can effectively represent dissimilar regions of ship images for each ship category. The diversity loss function $\mathcal{L}_{\text{div}}$ is defined as follows:

$$\mathcal{L}_{\text{div}} = \sum_{c=1}^{C} \left\|\mathbf{v}^{(c)}\mathbf{v}^{(c)T} - \mathbf{I}\right\|_F^2 \tag{8}$$

where $\mathbf{I} \in \mathbb{R}^{M\times M}$ denotes the identity matrix of size $M \times M$. $\|\cdot\|_F^2$ denotes the Frobenius norm. To ensure that the basis visual concepts of each category learn meaningful semantic information in the feature space, we subsequently introduce a constraint with a clustering and separation loss function $\mathcal{L}_{\text{cs}}$. This constraint aims to ensure that visual concepts of the same category cluster together, while those of different categories exhibit significant separation. This can be expressed as follows:

$$\mathcal{L}_{\text{cs}} = -\sum_{c=1}^{C} \frac{S_c^{\text{within}}}{S_c^{\text{all}}} \tag{9}$$

where S_c^{within} and S_c^{all} denote the correlation of intraclass and interclass basis visual concepts, respectively, which is

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given by

$$S_c^{\text{within}} = \sum_{\mathbf{v}_i^{(c)}, \mathbf{v}_j^{(c)} \in \mathbf{v}^{(c)}} s_{ij} = \sum_{\mathbf{v}_i^{(c)}, \mathbf{v}_j^{(c)} \in \mathbf{v}^{(c)}} \text{cor}\left(\mathbf{v}_i^{(c)}, \mathbf{v}_j^{(c)}\right) \tag{10}$$

$$S_c^{\text{all}} = \sum_{\mathbf{v}_i^{(c)} \in \mathbf{v}^{(c)}, \mathbf{v}_j \in V} s_{ij} = \sum_{\mathbf{v}_i^{(c)} \in \mathbf{v}^{(c)}, \mathbf{v}_j \in V} \text{cor}\left(\mathbf{v}_i^{(c)}, \mathbf{v}_j\right).i \tag{11}$$

D. Optimization Strategy

After VCL, we generate the feature vector $\mathbf{T}_{\text{out}}^{(c)} \in \mathbb{R}^M$, comprising the correlation scores between each multi-grained feature and the basis visual concept of the category c. This can be represented as $\mathbf{T}_{\text{out}}^{(c)} = [\text{cor}(\mathbf{F}_{f1}^{(c)}, \mathbf{v}_1^{(c)}), \text{cor}(\mathbf{F}_{f2}^{(c)}, \mathbf{v}_2^{(c)}), \ldots, \text{cor}(\mathbf{F}_{\text{fM}}^{(c)}, \mathbf{v}_M^{(c)})]^{\text{T}}$.

In the training phase, our initial approach involves the adoption of the softmax loss function for network optimization. While this strategy performs well in classification tasks, the discriminative capability of the learned features proves insufficient, resulting in inferior performance in retrieval tasks. Then, we resort to metric learning loss functions, including center loss [56], triplet loss [57], and contrastive loss [58]. However, these loss functions expose the network to challenges, such as sample batch combination explosion and difficulty in sample mining. The implementation of sophisticated sampling strategies complicates model training. To optimize the network that facilitates both image classification and retrieval tasks, we employ SphereFace [15] as the optimization objective for feature learning. The prediction score $p(y_i^{(c)})$ can be computed as follows:

$$p\left(y_i^{(c)}\right) = \frac{e^{\left\|\mathbf{T}_{\text{out}}^{(c)}\right\| \psi\left(\theta_{y_i^{(c)},i}\right)}}{e^{\left\|\mathbf{T}_{\text{out}}^{(c)}\right\| \psi\left(\theta_{y_i^{(c)},i}\right)} + \sum_{j \neq y_i^{(c)}} e^{\left\|\mathbf{T}_{\text{out}}^{(c)}\right\| \psi\left(\theta_{j,i}\right)}} \tag{12}$$

where $y_i^{(c)}$ indicate the label of ith ship image $\mathbf{x}_i$ in ship category c. $\psi(\theta_{y_i^{(c)},i})$ is angle function, which can be formulated as

$$\psi\left(\theta_{y_i^{(c)},i}\right) = (-1)^k \cos\left(m\theta_{y_i^{(c)},i}\right) - 2k \tag{13}$$

where $\theta_{y_i^{(c)},i} \in [(k\pi/m), (((k+1)\pi)/m)]$ and $k \in [0, m-1]$. $\theta_{y_i^{(c)},i}$ denote the angle between the weight vector and the feature vector $\mathbf{T}_{\text{out}}^{(c)}$ of category c. m is an integer controlling the size of angular margin. A larger m results in a more discriminative distribution of images across different ship categories on the sphere and a larger angular margin. Simultaneously, as m increases, ship images belonging to the same category become more distinct. To address imbalanced samples in the ship dataset, we employ focal loss [16] instead of cross-entropy loss as the task loss function $\mathcal{L}_{\text{task}}$, which can be defined as

$$\begin{aligned}\mathcal{L}_{\text{task}} = &-\sum_i \left(1 - p\left(y_i^{(c)}\right)\right)^\gamma y_i^c \log\left(p\left(y_i^{(c)}\right)\right) \\ &-\sum_i \left(p\left(y_i^{(c)}\right)\right)^\gamma \left(1 - y_i^c\right) \log\left(1 - p\left(y_i^{(c)}\right)\right).\end{aligned} \tag{14}$$

Finally, we jointly optimize the total loss function $\mathcal{L}_{\text{total}}$ in (15), which comprises causal local loss L_{cau}, concept diversity loss $\mathcal{L}_{\text{div}}$, clustering and separation loss $\mathcal{L}_{\text{cs}}$, and task loss $\mathcal{L}_{\text{task}}$

$$\mathcal{L}_{\text{total}} = \lambda_{\text{cau}}\mathcal{L}_{\text{cau}} + \lambda_{\text{div}}\mathcal{L}_{\text{div}} + \lambda_{\text{cs}}\mathcal{L}_{\text{cs}} + \mathcal{L}_{\text{task}} \tag{15}$$

where λ_{cau}, λ_{div}, and λ_{cs} denote three hyperparameters balancing the importance of different losses.

IV. Experiments and Analysis

A. Datasets Description

There are only two public datasets available for fine-grained ship classification in remote sensing images: FGSC-23 [4] and FGSCR-42 [5]. During the experimental process, we observe that the categorization granularity of ship images in the FGSC-23 is insufficient. For example, there is no further subdivision based on specific destroyer models. In the case of FGSCR-42, some images are partially cropped, resulting in incomplete ship outlines [Fig. 7(a)]. Additionally, some ship images are not captured by optical satellites [Fig. 7(b)]. To better align with real-world application scenarios and more effectively evaluate the superiority and interpretability of our model in fine-grained ship image classification and retrieval tasks, we construct the FGSRSI-23 dataset.

1) *FGSC-23 [4]:* FGSC-23 is the first fine-grained ship classification dataset in remote sensing images that we constructed previously. Comprising 23 ship categories, it includes a total of 4080 remote sensing image chips sourced from Google Earth and the GF-1 satellite, with resolutions ranging from 0.4 to 2 m. Each ship image is labeled with a ship category, aspect ratio, and distribution direction, with image sizes spanning from 40 to 800 pixels.
2) *FGSCR-42 [5]:* FGSCR-42 is a large fine-grained ship classification dataset in remote sensing images, encompassing 9320 image chips across 42 fine-grained ship categories. Collected from Google Earth and various public remote sensing image datasets, the image sizes range from 50 × 50 to approximately 500 × 1500 pixels.
3) *FGSRSIS-23:* FGSRSIS-23 is the first fine-grained ship dataset in remote sensing image slices. These ship image slices are mainly collected from SuperView-1 (SV-1) satellite and Google Earth. FGSRSIS-23 comprises of a total 1067 image slices and all the image slices are annotated by professionals. The dataset is divided into nine primary ship categories and 23 subcategories, including Arleigh Burke-class Destroyer (ABD), ATD, Asagiri-class Destroyer (ASD), Izumo-class helicopter Destroyer (IHD), Murasame-class Destroyer (MUD), Avenger-class Mine Countermeasures Ship (AMCS), Takanami-class Destroyer (TAD), Wasp-class Amphibious Assault Ship (WAAS), Ferry-class Dock Landing Ship (FDLS), Endurance-class Tank Landing Ships (ETLSs), Kongō-class Destroyer (KOD), Formidable-class Frigate (FOF), Blue Ridge-class Command Ship (BRCS), Nimitz-class Aircraft Carrier (NAC), Oliver Hazard Perry-class Frigate (OHPF), USS Enterprise (USS), Akitsuki-class Destroyer (AKD), Hyuga-class Helicopter Destroyer (HHD), San Antonio-class

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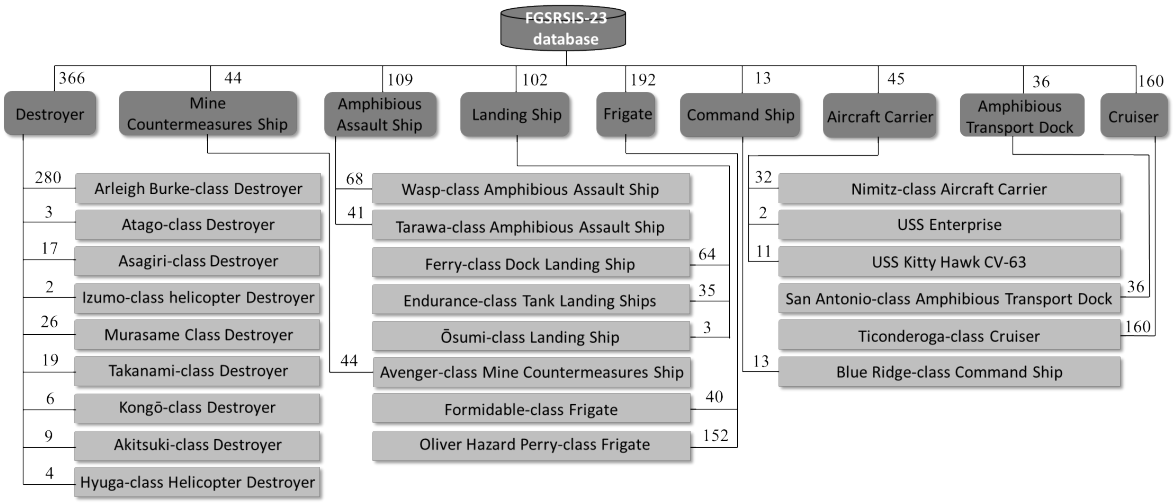

Fig. 5. Taxonomy of FGSRSIS.

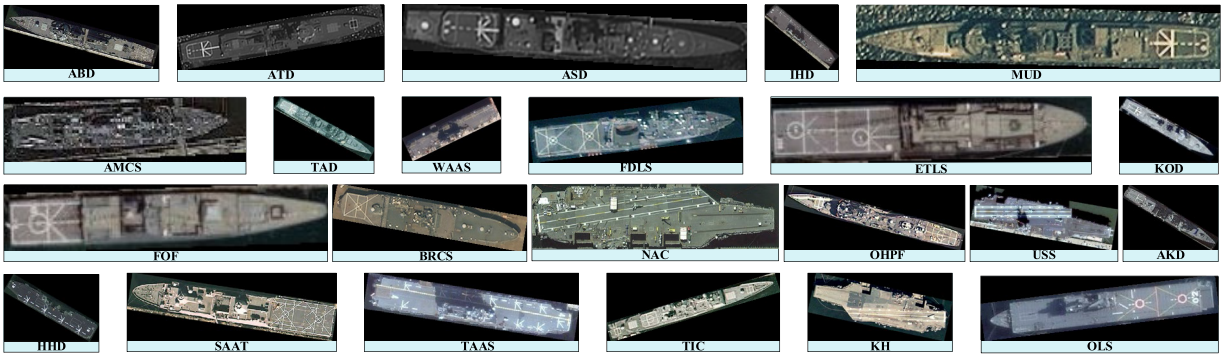

Fig. 6. Some ship image samples from FGSRSIS-23.

Amphibious Transport Dock (SAAT), Tarawa-class Amphibious Assault Ship (TAAS), Ticonderoga-class Cruiser (TIC), USS Kitty Hawk CV-63 (KH), and Ōsumi-class Landing Ship (OLS). The sizes of ship image slices vary, ranging approximately from 30 to 800 pixels, with resolutions spanning from 0.4 to 2 m. These image slices encompass diverse illumination conditions and exhibit arbitrary-oriented distributions. The categories in FGSRSIS-23 are imbalanced, mirroring real-world scenarios. The taxonomy of FGSRSIS is shown in Fig. 5. Compared with FGSC-23 and FGSCR-42, FGSRSIS offers several advantages: 1) the ship images of the same category are further divided, posing greater challenges for fine-grained image classification. 2) Each image has only one ship target and the integrity of the ship target contour in the image is preserved. 3) All the ship images only capture slices of the ship's target, excluding background interference, so that researchers can better focus on the fine-grained ship classification task. Some ship image samples from FGSRSIS-23 are displayed in Fig. 6. The dataset is available at https://github.com/xiong577.

B. Experimental Setup

In our experiment, we utilize two publicly available fine-grained datasets, FGSC-23 and FGSCR-42, along with our constructed dataset, FGSRSIS-23, to assess the performance of the proposed Cog-net. Specifically, FGSCR-42 is randomly split into a training subset and a testing subset in a 1:1 ratio. For the FGSC-23 and FGSRSIS-23 datasets, 80% of ship images in each category are allocated for training, with the remainder reserved for testing. During the data preprocessing stage, all images are resized to 224×224 pixels, incorporating zero-padding operations to maintain aspect ratios. Various data augmentation techniques, including rotation and image scaling, are employed to balance the image samples in the training stage. Ship images from the training subset undergo rotation up to 15° and scaling up or down by 20% of the image size. To extract high-level features from the input ship images, several mainstream convolutional backbone models (pretrained on ImageNet [59]) are utilized, including VGG16 [60], ResNet18 [61], ResNet50 [61], and DenseNet-121 [62].

In the training stage, each basis visual concept is trainable model parameters initialized with random values drawn from a uniform distribution within the range $[0, 1]$. Following the training strategy of [36], [37], and [38], we employ a fixed FC layer initialized with $+1.0$ and -0.5 for correct and incorrect connection weights in the first stage to optimize the concepts and the parameters preceding the last layer. Subsequently, each visual concept is updated with its nearest latent vectors to ensure the concepts can be visualized as examples within the training distribution. Finally, with the learned parameters and fixed concepts, the weights in the final predicted layer are

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optimized to elucidate the contributions of each concept to the decision-making process. All models and datasets undergo training for 80 epochs using the stochastic gradient descent (SGD) optimizer with mini-batch, initialized at a learning rate of 0.001. We set the number of attention map M to 10, representing the number of basis visual concepts learned for each ship category. In the total loss function, λ_{cau} and λ_{cs} are set to 0.5, and λ_{div} is set to 0.01. The balance parameter γ in the local loss function is set to 2. The size of the angular margin m in optimization objection is set to 4. The experiments are conducted using the PyTorch deep learning framework, executed on Ubuntu 16.04, with 32 GB of RAM, eight Intel Core i7-6770K CPU, and NVIDIA RTX 2080Ti.

For the image classification task, we evaluate the proposed model's effectiveness using overall accuracy (OA) and a confusion matrix (CM). OA quantifies the proportion of accurately predicted labels for all images in the testing set. CM graphically represents the ratio of correctly predicted labels for a specific category in the testing set. In the image retrieval task, the performance of the proposed model is assessed using mean average precision (mAP) and precision at k samples (P@k). Specifically, mAP considers the order of all ground truths in the retrieved ship images during the search process, providing a comprehensive evaluation of the ship retrieval task's performance. P@k indicates the precision of the top k retrieved images.

C. Qualitative Results

We visualize the basis visual concepts generated from the training images and the corresponding ship regions from the test images. As shown in Fig. 8, two basis visual concepts for a specific category from three datasets are randomly selected. We adopt an activation map to visualize the regions represented by basis visual concepts in ship images. For each concept, we list four corresponding activation maps generated from test images. From the visualization results, each basis visual concept captures fine-grained and discriminative ship features for each category. The heatmaps show that the same ship region is activated in different images and corresponds to the regions of the respective basis visual concept. For instance, in the FGSRSIS-23 dataset, the *afterdeck* concept, as shown in Fig. 8(a), and the *midship* concept, as shown in Fig. 8(b), for the OHPF category are sufficiently represented, and the activated ship regions in the four test images align with these concepts. In the basis visual concept column, different concept mainly captures diverse visual evidence from a human perspective. In the four test images column, there is minimal overlap between the activated regions of different basis visual concepts within the same ship category. The visualization results effectively demonstrate the diversity and effectiveness of the basis visual concepts learned by the network.

The interpretable reasoning process of our Cog-Net for fine-grained ship image classification task is illustrated in Fig. 9. Each reasoning branch calculates the correlation score between key regions of test images, which are activated, and the corresponding basis visual concept generated from the training images. These correlation scores are then combined to determine the contribution score and make predictions. Specifically,

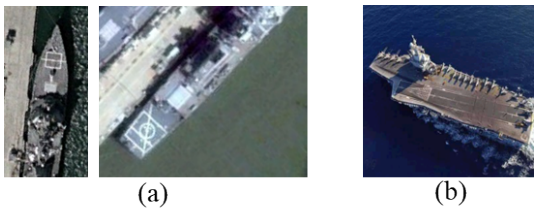

Fig. 7. Examples of ship images from FGSCR-42 dataset. (a) Incomplete outlines of ship images. (b) Ship image is not captured by optical satellites.

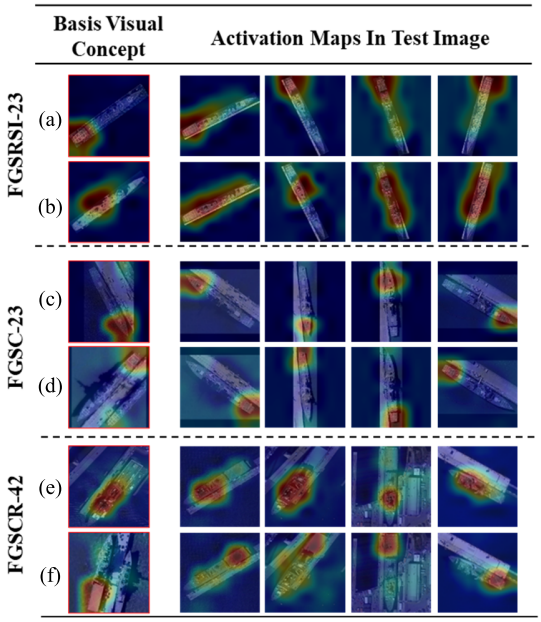

Fig. 8. Visualization results of the basis visual concept from the training images and their corresponding activation maps from test images. (a) *Afterdeck* concept of OHPF category from FGSRSIS-23 dataset. (b) *Midship* concept of OHPF category from FGSRSIS-23 dataset. (c) *Foredeck* concept of Landing Craft category from FGSC-23 dataset. (d) *Stern* concept of Landing Craft category from FGSC-23 dataset. (e) *Superstructure* concept of Whitby Island-class Dock Landing Ship category from FGSCR-42 dataset. (f) *Afterdeck* concept of Whitby Island-class Dock Landing Ship category from FGSCR-42 dataset.

when classifying an Endurance-class Tank Landing Ship from the FGSRSIS-23 dataset [Fig. 9(a)], our Cog-Net successfully captures fine-grained features semantically close to some basis visual concepts such as Endurance-class Tank Landing ship's *afterdeck*, *superstructure*, and *bow*. By calculating the correlation score and obtaining the contribution score for each reasoning branch, we accumulate a relatively high total points (7.413) for the category Endurance-class Tank Landing Ship. Thus, the visualization results of the reasoning process demonstrate that the proposed Cog-Net provides interpretable evidence for the final predicted results in a manner consistent with human cognition.

For the fine-grained ship image retrieval task, the interpretable reasoning process of our Cog-Net is shown in Fig. 10. The leftmost image in each category serves as the query ship image, while the rightmost one represents the ship image from the gallery. By calculating the correlation scores between key regions in query ship images and the basis visual concepts generated from training images, as well as the correlation scores between key regions in gallery ship images and the

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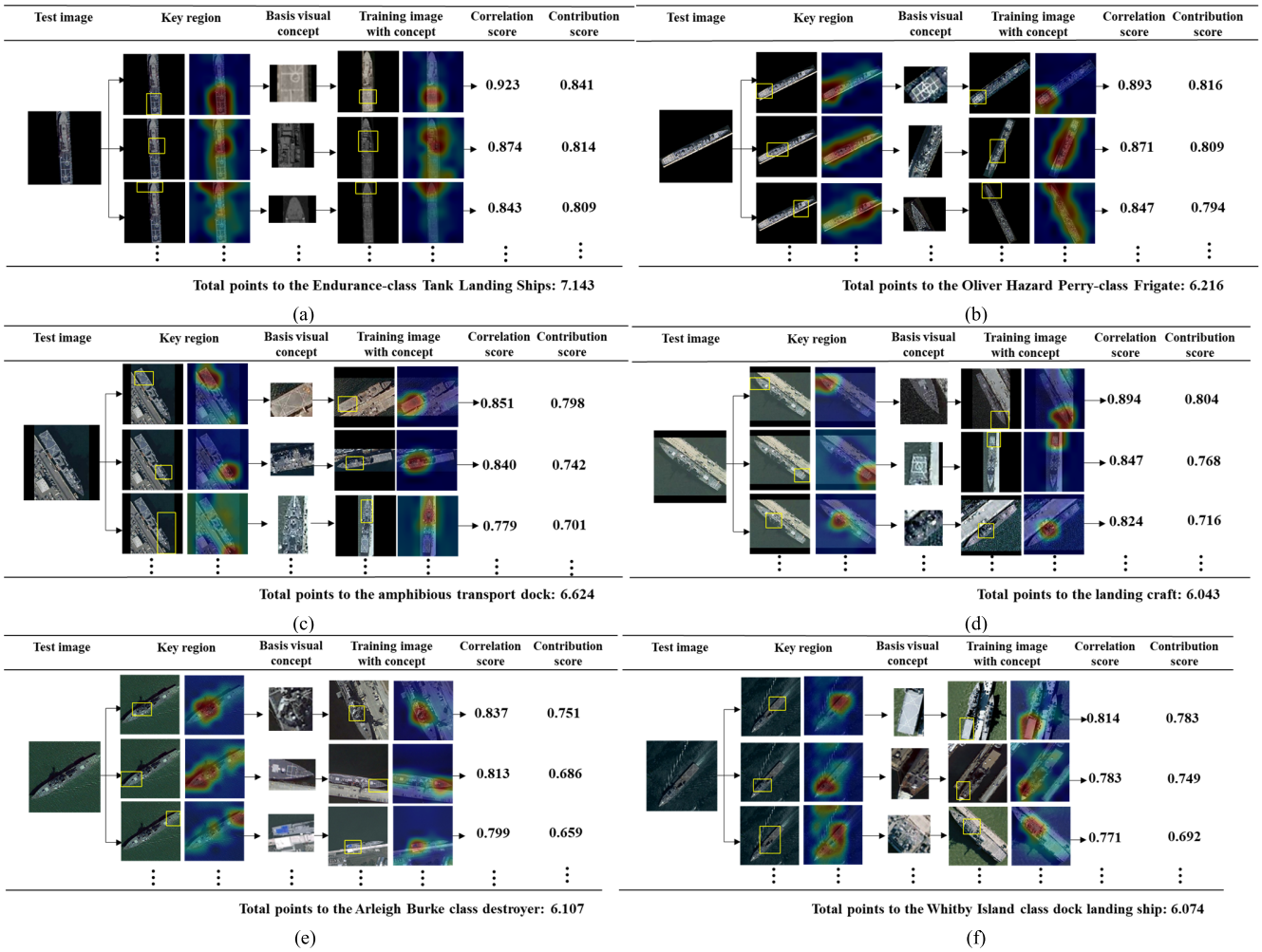

Fig. 9. Visualization of our Cog-Net reasoning process for fine-grained ship image classification task. (a) Visual reasoning process in classifying Endurance-class Tank Landing Ship category from FGSRSIS-23 dataset. (b) Visual reasoning process in classifying OHPF category from FGSRSIS-23 dataset. (c) Visual reasoning process in classifying Amphibious Transport Dock category from FGSC-23 dataset. (d) Visual reasoning process in classifying Landing Craft category from FGSC-23 dataset. (e) Visual reasoning process in classifying ABD category from FGSCR-42 dataset. (f) Visual reasoning process in classifying Whitby Island-class Dock landing Ship category from FGSCR-42 dataset.

basis visual concepts generated from training images, our proposed Cog-Net can find the most semantically similar ship image to the query ship image from the gallery. Specifically, given a query image from the Medical Ship category in the FGSCR-42 dataset [Fig. 10(c)], our Cog-Net finds several basis visual concepts, such as Medical Ship's *deck*, *red cross*, and *bow*, existed in the query ship image by calculating the correlation score between each key region and basis visual concept. Similarly, we calculate the correlation scores to find gallery images with these concepts. Similar to the classification task, we generate the contribution score of each branch and sum them to obtain the total points for query and gallery images, respectively. The total correlation score is obtained by multiplying the total score of query and gallery images. Subsequently, we can find Medical Ship images from the gallery that are similar to the query image and rank them by the total correlation score in the final retrieval result.

D. Ablation Study

In this section, we conduct extensive ablation studies to illustrate the performance of Cog-net using various convolutional backbones. We also assess the effectiveness of its different components. Additionally, we explore the impact of the number of concepts and analyze the network's complexity.

1) Cog-Net With Different Convolutional Backbones: To comprehensively assess the performance of our Cog-Net for ship image classification and retrieval, we conduct an ablation study on three datasets. For the image classification task, we first adopt OA as the evaluation metric to quantitively verify the effectiveness of Cog-Net. The experimental results with different convolutional backbones are presented in Table I. It is evident that, as the network depth increases, the accuracy gradually improves, with DenseNet-121 achieving the best classification results across all datasets. All backbones demonstrate classification accuracy exceeding 90% on the FGSC-23 and FGSCR-42 datasets and over 80% on the FGSRSIS-23 dataset. Similar to the classification task, we conduct an ablation study on three datasets to verify the effectiveness of our Cog-Net on the fine-grained ship image retrieval task. The experimental results in Table IV show that Cog-Net achieves high accuracy in retrieval tasks using different backbones. Particularly noteworthy is the outstanding performance of DenseNet-121, which achieves a P@1 value greater than 90% and a mAP value greater than 80% on three datasets. This

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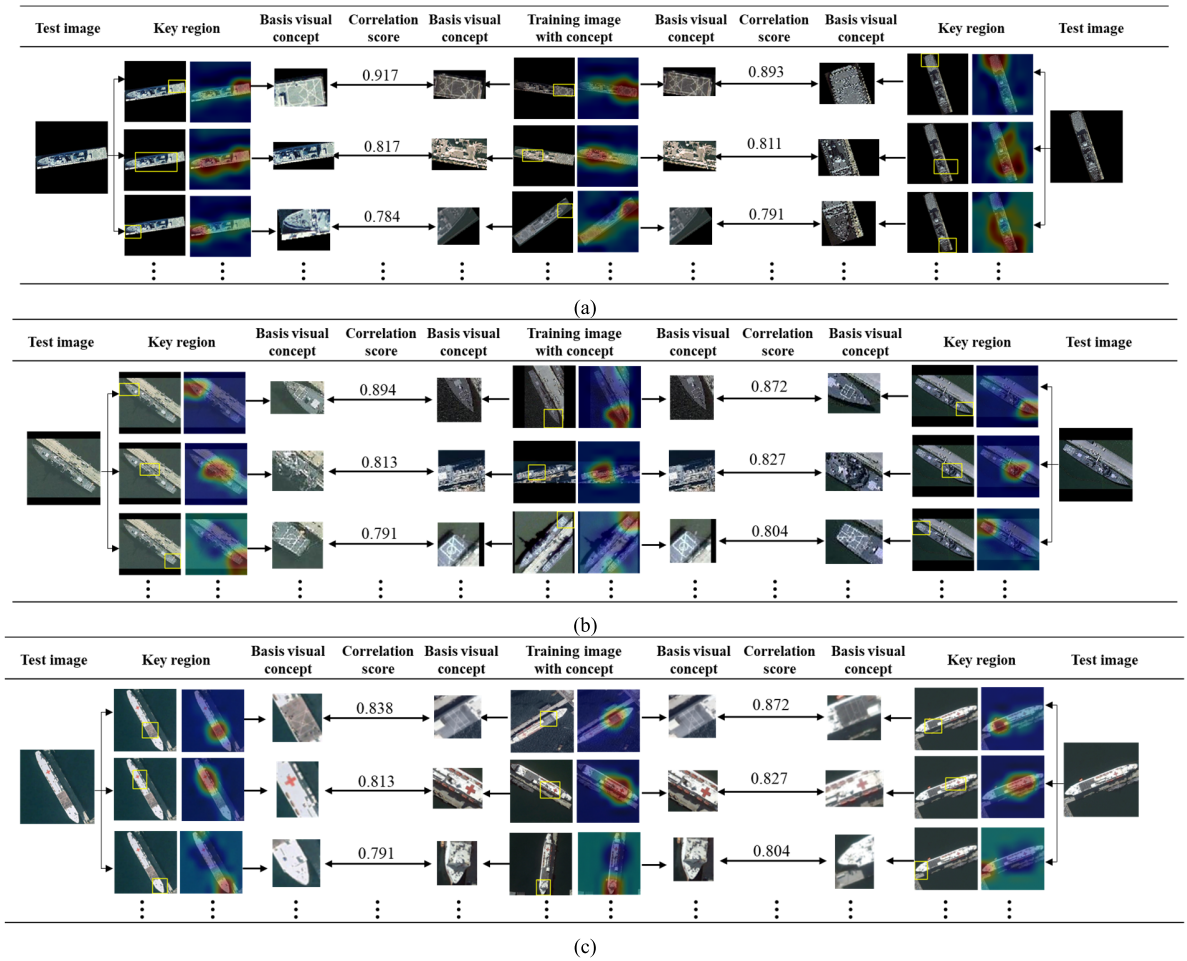

Fig. 10. Visualization of our Cog-Net reasoning process for fine-grained ship image retrieval task. (a) Visual reasoning process of retrieval task for SAAT category from FGSRSIS-23 dataset. (b) Visual reasoning process of retrieval task for Landing Craft category from FGSC-23 dataset. (c) Visual reasoning process of retrieval task for Medical Ship category from FGSCR-42 dataset.

TABLE I

CLASSIFICATION ACCURACIES OF THE PROPOSED COG-NET UNDER DIFFERENT CONVOLUTIONAL BACKBONES

Backbone	FGSC-23	FGSCR-42	FGSRSIS-23	Drop
VGG16	90.42	95.87	82.97	3.01
ResNet18	91.87	97.01	85.23	1.73
ResNet50	92.19	97.89	85.71	1.17
DenseNet-121	**93.02**	**98.09**	**86.18**	-

TABLE II

CLASSIFICATION ACCURACIES OF THE PROPOSED COG-NET UNDER DIFFERENT SETTINGS

Setting	FGSC-23	FGSCR-42	FGSRSIS-23	Drop
w/o $L_{\overline{cau}}$	87.91	89.13	81.28	6.33
w/o $L_{\overline{div}}$	91.49	94.37	84.61	2.28
w/o L_{cs}	88.42	90.93	82.03	5.31
All	**93.02**	**98.09**	**86.18**	-

demonstrates the robustness and effectiveness of Cog-Net in the ship image retrieval task.

2) Effectiveness of the Different Components: Subsequently, we conduct an ablation study with four variations of our Cog-Net: 1) removing the causal local loss $\mathcal{L}_{\text{cau}}$ (generating multiple attention features without a causal counterfactual strategy); 2) removing the concept diversity loss $\mathcal{L}_{\text{div}}$; 3) removing the clustering and separation loss $\mathcal{L}_{\text{cs}}$; 4) the proposed Cog-Net with all components. The experimental results in Table II show that removing any of the losses in Cog-Net results in varying degrees of decrease in OA values. Specifically, removing $\mathcal{L}_{\text{cs}}$ results in a 5.31% reduction. This demonstrates that the learned concepts are less constrained without $\mathcal{L}_{\text{cs}}$ and unbale to generate some specific discriminative basis visual concept for each ship category. Consequently, in the VCL strategy, incorporating $\mathcal{L}_{\text{cs}}$ is essential to generate several category-specific basis visual concepts. These concepts play a vital role in fully representing the distinctive characteristics of each ship category, thereby enhancing their utility for fine-grained classification. Removing $\mathcal{L}_{\text{cau}}$ causes a 6.33% drop in OA value, attributed to a spurious correlation between learned concepts and final prediction result, resulting in inaccurate ship parts localization and unreliable predictions. Constrained by $\mathcal{L}_{\text{cau}}$, the network can extract multigrained discriminative features of objects and establish genuine causal

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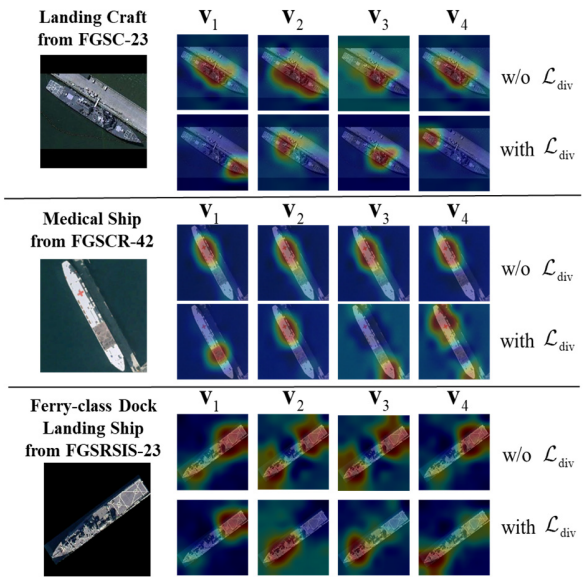

Fig. 11. Visualization results of basis visual concepts from the training images and their corresponding activation maps from test images.

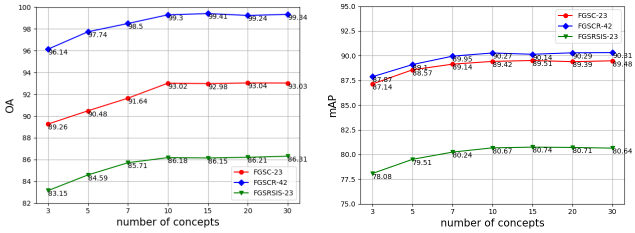

Fig. 12. Curves of OA and mAP values with respect to different numbers of concepts.

TABLE III
COMPARISON OF PARAMETERS AND FLOPS OF BACKBONES AND COG-NET

Method	Params(M)	FLOPs(G)
VGG16	14.11	15.34
Cog-Net (VGG16)	14.80	15.35
ResNet18	10.69	1.82
Cog-Net (ResNet18)	11.26	1.83
ResNet50	23.16	4.11
Cog-Net (Resnet50)	23.79	4.14
DenseNet-121	6.08	2.88
Cog-Net (DenseNet-121)	7.11	2.90

relationships between the learned concepts and classification outcomes. When removing $\mathcal{L}_{\text{div}}$ from Cog-Net, the value of OA drops by 2.28%. This is mainly because the learned basis visual concepts fail to capture diverse ship's fine-grained parts and generate some redundant concepts without the supervision of $\mathcal{L}_{\text{div}}$. Supplemented by $\mathcal{L}_{\text{div}}$, we can guide the network to explore multiple visual concepts, ensuring that distinct basis vectors represent different concepts within a given ship category. We further conduct the visualization experiments to validate the importance of $\mathcal{L}_{\text{div}}$ in our Cog-Net. As shown in Fig. 11, without adopting $\mathcal{L}_{\text{div}}$, the ship parts of concern by Cog-Net are quite limited, with some overlap between different concepts. In comparison, diverse discriminative parts of the ship image, such as the ship's *deck*, *red cross*, *and bow* are fully represented with $\mathcal{L}_{\text{div}}$ applied. The visualization results further verify $\mathcal{L}_{\text{div}}$ is essential to provide an interpretable visual reasoning process for the predicted results. To better illustrate the classification accuracy for each ship category, we adopt the CM to evaluate the performances of four variations of our Cog-Net on three datasets. As shown in Fig. 13, the diagonal values of CM represent the accuracy rate (AR) for each category, while the other values denote the misclassification rate (MR) for each category. We observe that discarding any loss function of Cog-Net results in an increased MR for some subcategories. The proposed Cog-Net, combining all components, obtains the best AR for most of the ship sub-categories on the three datasets, further verifying the effectiveness of our Cog-Net.

For the image retrieval task, we also conduct an ablation study with four variations of our Cog-Net in Table V. From the experimental results, we can see that the causal counterfactual strategy guides the network to explore the intrinsic relationship between several ship concepts and predicted outcomes, improving the values of P@1 and mAP by 6.49% and 8.21%, respectively. The introduction of $\mathcal{L}_{\text{div}}$ slightly improve the performance of Cog-Net, resulting in an increment of 1.04% on the P@1 value and 1.00% on the mAP value. This improvement is attributed to the diversity loss, which aids Cog-Net in concentrating on various ship regions and mitigating the acquisition of redundant concepts. $\mathcal{L}_{\text{cs}}$ is essential for Cog-Net to learn some discriminative basis visual concepts and use them for ship image retrieval, offering a 4.96% improvement in P@1 value and a 4.73% improvement in mAP value, respectively.

3) Impact of the Number of Concepts: We perform the ablation studies on three datasets to illustrate the impact of the different number of concepts on classification and retrieval performance. As depicted in Fig. 12, when the number of concepts is below ten, both OA and mAP values exhibit a gradual increase with the rising number of concepts. When the number exceeds 10, the variations in OA and mAP become relatively negligible, reaching a plateau despite further increases in the number of concepts. Acknowledging that an escalation in concepts contributes to augmented model parameters and the generation of redundant concepts, we set the number of concepts to 10 in our experiment.

4) Parameters and FLOPs: To assess the complexity of our Cog-Net, we present a summary of the parameters (Params) and floating point operations (FLOPs) for Cog-Net and various backbones (VGG16, ResNet18, ResNet50, and DenseNet-121). Table III reveals that the parameters and FLOPs of Cog-Net are essentially equivalent to those of the diverse backbones. The supplementary parameters and FLOPs introduced by Cog-Net prove negligible in comparison with the backbones. Despite these marginal increments in model parameters and computational costs, our approach not only enhances classification and retrieval accuracies but also furnishes an interpretable reasoning process for the networks.

E. Comparison With Other Methods

To demonstrate the superiority of our Cog-Net, we conduct a quantitative assessment of classification results

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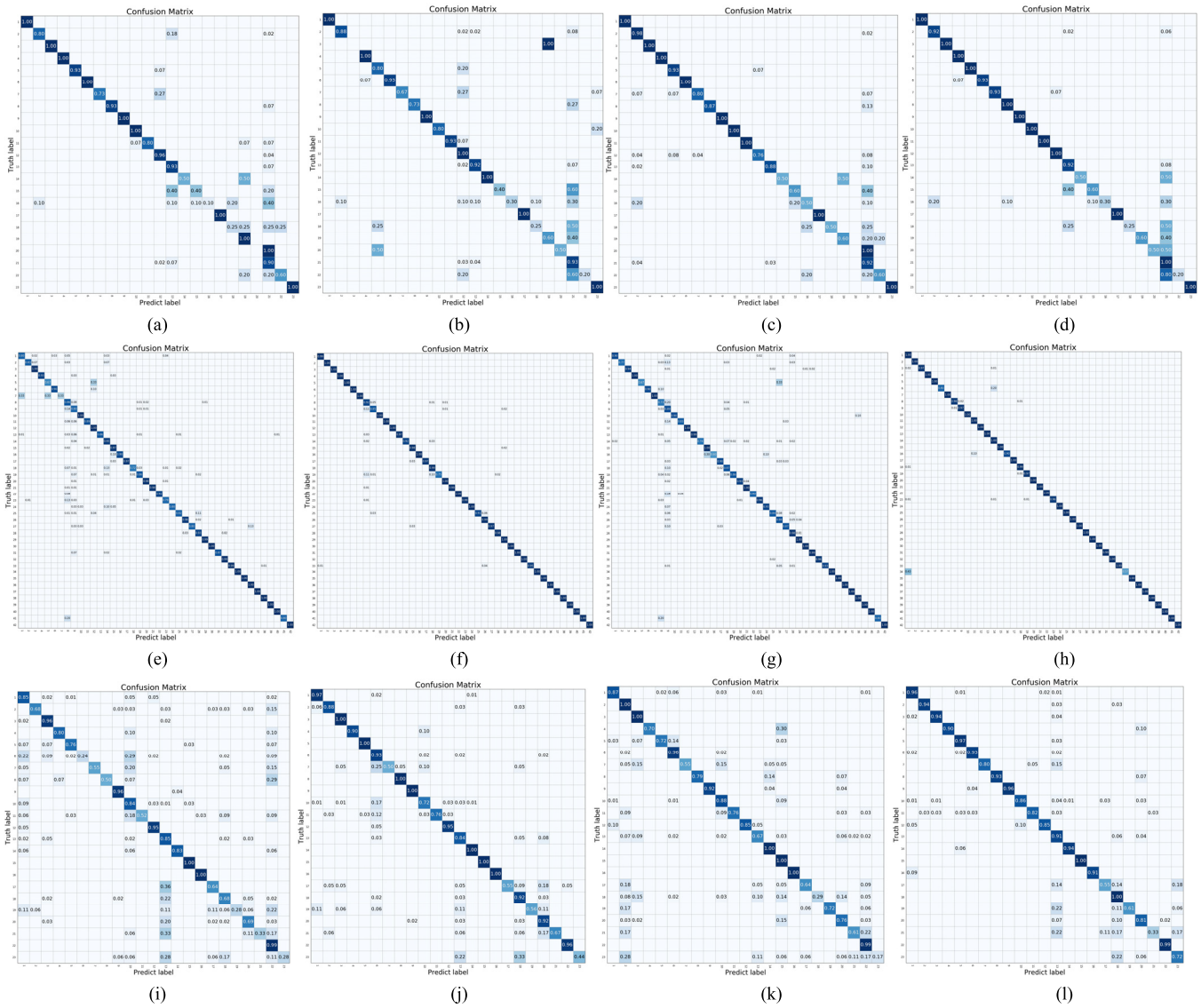

Fig. 13. CM of four variations of our Cog-Net on three datasets. (a)–(d) Results of four variations of Cog-Net on dataset FGSC-23. (e)–(h) Results of four variations of Cog-Net on dataset FGSCR-42. (i)–(l) Results of four variations of Cog-Net on dataset FGSRSIS-23. (a), (e), and (i) w/o $\mathcal{L}_{\text{cau}}$. (b), (f), and (j) w/o $\mathcal{L}_{\text{div}}$. (c), (g), and (k) w/o $\mathcal{L}_{\text{cs}}$. (d), (h), and (l) All.

TABLE IV
RETRIEVAL ACCURACIES OF THE PROPOSED COG-NET UNDER DIFFERENT CONVOLUTIONAL BACKBONES

Backbone	FGSC-23		FGSCR-42		FGSRSIS-23		Drop	
	P@1	mAP	P@1	mAP	P@1	mAP	P@1	mAP
VGG16	90.28	80.76	92.27	83.17	84.68	74.83	5.67	7.20
ResNet18	93.08	87.08	96.13	88.97	88.12	76.97	2.31	2.45
ResNet50	93.97	89.31	96.24	89.01	88.15	78.88	1.96	1.05
DenseNet-121	**95.38**	**89.42**	**97.98**	**90.27**	**90.89**	**80.67**	-	-

obtained by different classification approaches. The comparative experimental results on three datasets are reported in Table VI. For the FGSC-23 dataset, ME-CNN [63] aggregates course-grained features of the ship, which lack discriminative capabilities for representing ship subcategories. SIM [64], IELT [65], and DCN [66] learn fine-grained feature representations for natural images but prove less effective for this remote sensing ship image dataset. LGFFE [17] explores contextual information in remote sensing scene images, generating multi-grained features by fusing global and local features. P2Net [6] leverages an asynchronous contrastive learning strategy to tackle the imbalanced fineness of ship subcategories, but the vision transformer backbone fails to perform well on the FGSC-23 dataset. AMEFRN(sc2) [4] extracts multilevel image features, capturing fine-grained information in remote sensing ship images. Recently proposed EA-Net [12], C2NET [67],

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TABLE V
RETRIEVAL ACCURACIES OF THE PROPOSED COG-NET UNDER DIFFERENT SETTINGS

Setting	FGSC-23		FGSCR-42		FGSRSIS-23		Drop	
	P@1	mAP	P@1	mAP	P@1	mAP	P@1	mAP
w/o $\mathcal{L}_{cau}$	88.29	81.38	90.01	78.29	86.48	76.07	6.49	8.21
w/o $\mathcal{L}_{div}$	94.24	89.08	96.82	90.05	90.06	78.24	1.04	1.00
w/o $\mathcal{L}_{cs}$	91.23	85.79	92.08	86.29	86.07	74.08	4.96	4.73
All	**95.38**	**89.42**	**97.98**	**90.27**	**90.89**	**80.67**	-	-

TABLE VI
COMPARISON OF CLASSIFICATION ACCURACY OF DIFFERENT METHODS

Dataset	Model	OA
FGSC-23	ME-CNN [63]	85.58
	SIM [64]	86.30
	P2Net [6]	88.99
	LGFFE [17]	89.45
	DCN [66]	90.66
	IELT [65]	91.08
	AMEFRN(sc2) [4]	91.15
	EA-Net [12]	91.78
	C2NET [67]	92.48
	EFM-Net [7]	92.60
	Cog-Net	**93.02**
FGSCR-42	DCN [66]	93.03
	P2Net [6]	93.21
	TASN [68]	93.51
	EA-Net [12]	94.06
	C2NET [67]	95.42
	IELT [65]	97.31
	FFVT [69]	97.90
	SIM [64]	97.90
	PMG-V2 [70]	98.00
	EFM-Net [7]	**99.30**
	Cog-Net	98.09
FGSRSIS-23	ME-CNN [63]	76.28
	MCNN [71]	76.91
	LGFFE [17]	79.24
	DCN [66]	80.09
	IELT [65]	81.97
	CMN [72]	83.02
	EFM-Net [7]	83.41
	EA-Net [12]	84.28
	SR-GNN [73]	84.75
	Cog-Net	**86.18**

and EFM-Net [7] focus on learning some essential features of different objects in remote sensing images, achieving satisfactory results for fine-grained classification tasks. The proposed Cog-Net attains the best results, with OA values increased by 0.42% and 1.24% compared with the second-ranked EFM-Net [7] and the third-ranked EA-Net [12], respectively. For the FGSCR-42 dataset, our Cog-Net achieves competitive classification performance, with a classification accuracy of 98.09%, only experiencing a decrease of less than 1% compared with Cog-Net, which achieves the best result. EFM-Net utilizes both the feature mining module and data augmentation module to investigate discriminative image regions, achieving optimal classification performance. It is essential to clarify that our Cog-Net aims not only to achieve superior classification results but also to provide a comprehensible reasoning process akin to human understanding. We incorporate only a subset of standard data augmentation techniques to enhance performance, acknowledging the potential for further accuracy improvement with the utilization of additional techniques. For the FGSRSIS-23 dataset, our Cog-Net surpasses all compared approaches, achieving a new state-of-the-art classification result and presenting a significant improvement of 1.43% compared with the second-ranked SR-GNN [73]. The comparative results on the three datasets verify the advancement of our Cog-Net in tackling fine-grained remote sensing ship image classification tasks. i For the fine-grained ship image retrieval task, we conduct a comparison experiment on three datasets in Table VII. FAH [27], DHCNN [74], and AHCL [75] propose deep networks to extract high-level information and generate some binary hash codes to represent features of remote sensing images. However, these methods primarily focus on extracting global features from entire remote sensing scene images, proving less effective in distinguishing subtle visual differences between different ship subcategories and achieving fine-grained ship image retrieval. AMCL [25] combines channel and spatial attention strategies to extract salient features from remote sensing scene images but overlooks the tiny difference of ship subcategories. IFSN [13] and DBHT [29] aim to identify essential regions in remote sensing ship images and introduce specific constraints to enhance retrieval accuracy. Nevertheless, these methods are tailored for retrieving images from only six coarse-grained ship categories. Their effectiveness in extracting fine-grained features from diverse subcategories of ship images is limited. In the field of computer vision, ExchNet [76] and FISH [77] leverage attention strategies and constraints to learn discriminative hash codes for various subcategories, achieving retrieval success for natural images such as cars, aircraft, foods, and birds. While these hash learning-based methods expedite the retrieval process and reduce storage costs, the information loss during hash code generation may limit their performance in extracting rich features from complex remote sensing ship images. DGCRL [78], PCE [79], MGA [80], and PNCA++ [81] adopt metric learning strategies to improve the discriminative ability of feature representation. However, the notable differences between natural images and remote sensing ship images hinder

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TABLE VII
COMPARISON OF RETRIEVAL ACCURACY OF DIFFERENT METHODS

Method	FGSC-23		FGSCR-42		FGSRSIS-23	
	P@1	mAP	P@1	mAP	P@1	mAP
FAH [27]	85.09	74.14	83.17	75.19	74.14	66.17
DHCNN [74]	85.18	76.98	85.19	78.87	74.82	68.98
AHCL [75]	86.27	79.29	86.18	77.04	76.18	72.16
ExchNet [76]	88.14	81.18	87.14	79.18	80.28	75.85
FISH [77]	91.32	87.18	91.97	86.01	83.05	77.14
DBHT [29]	89.48	84.29	88.18	84.29	80.97	77.09
IFSN [13]	90.19	88.08	92.16	85.26	81.16	77.29
AMCL [25]	89.01	80.11	89.14	82.45	83.47	75.14
DGCRL [78]	92.28	86.24	91.28	86.21	84.84	77.05
MGA [80]	92.38	86.78	94.25	88.05	84.09	77.91
PCE [79]	93.17	87.97	95.28	88.16	85.16	76.98
PNCA++ [81]	94.98	88.46	97.54	88.59	86.08	76.62
Cog-Net	**95.38**	**89.42**	**97.98**	**90.27**	**90.89**	**80.67**

the direct application of these methods to the remote sensing ship image retrieval task. Our proposed Cog-Net integrates the local and global image features and explores intrinsic causal relationships between various discriminative ship regions and predictions, achieving superior retrieval accuracy on the three datasets. Notably, on our challenging FGSRSIS-23 dataset, our Cog-Net improves the value of P@1 and mAP by 4.81% and 4.05%, respectively, compared with the second-ranked PNCA++ [81]. This further demonstrates that Cog-Net not only provides a clear and interpretable reasoning process but also possesses powerful discriminative capabilities.

V. CONCLUSION

In this study, we propose Cog-Net, a novel framework designed for fine-grained ship classification and retrieval in remote sensing images, with the objective of imitating the decision-making process from primitive perception to advanced cognition. The network consists of a CMFL module and a VCL module. In the first stage, the network searches for salient regions within ship images to facilitate VCL. The causal local branch generates attention maps to represent different parts of ship images and incorporates a causal graph to lead the network to have counterfactual thinking like humans. In the second stage, the network learns basis visual concepts for each ship category, yielding predicted results through visual reasoning. We introduce a new dataset, FGSRSI-23, for a comprehensive evaluation. Experimental results demonstrate the superior performance of the proposed network over existing methodologies in ship image classification and retrieval tasks, providing transparent and interpretable reasoning processes for the predicted results.

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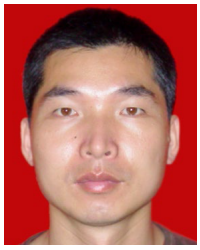

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